

landscapes but for everything else you have to get much closer to get the subject in the frame.

Most phones don't have optical zoom, so you will get a single focal length figure. Some, such as the Galaxy S4 Zoom, have two figures, indicating the focal length at minimum zoom (or wide mode) and focal length at maximum zoom (or tele mode). Zooming is essentially artificially increasing the focal length of the lens, which reduces the field of view, thereby giving the illusion of getting closer to the subject.

The other figure we mentioned earlier is lens speed. This number denotes the width that the aperture is capable of opening. On standard cameras with variable aperture, you will find two figures, one denoting the smallest aperture size and the other the maximum aperture size. Since mobile phone lenses usually have a fixed aperture size, you get just one figure, occasionally printed on the side of the lens like, say, $f/2.0$.

Aperture is measured in f-stops, and is often written as $f/1.8$, $f/2.9$, $f/2.4$, etc. The smaller the f-number, the wider the aperture is capable of opening and vice-versa. As you can imagine, an aperture that opens wide is preferable as it allows in more light for the same shutter speed compared to a narrower aperture. This means you can use a faster shutter speed with a lens with a wide aperture and still get the same amount of light as using a slow shutter speed with a narrow aperture. This is why lenses with a wide aperture are called 'fast' lenses.

Other than allowing more light in, a fast lens, i.e., one with a wide aperture, also has a shallower depth of field. This is not really apparent on a mobile phone camera as the lens is too small physically for this to be noticeable but if you've played around with DSLR lenses then you'd have noticed how fast lenses have a deliciously shallow depth of field compared to slow lenses. Depth of field is inversely proportional to the size of the aperture, so a wide aperture has less or shallow depth of field and vice-versa.

Mobile phone camera lenses are usually made of plastic but some manufacturers, such as Nokia, choose to go with higher quality branded lens from companies like Carl Zeiss who make glass lenses. A lens is the single most important part of the camera system as all the light passes through it. If the lens is bad then the image is affected negatively. A lens can make or break an image and having a good lens is of utmost importance. A good lens will be free of distortion, such as purple fringing, corner softness and barrel/pincushion distortion. However, advancement in camera software means that often the camera is able to fix some of the problems of a cheaper lens in post processing.